

Approximating Euler's number



- 1
- | | | | | | | | |
|---|---------|---|--------|---|--------|---|--------|
| a | 54.5982 | b | 0.3679 | c | 1.2214 | d | 8.2436 |
| e | 1.4715 | f | 0.0163 | | | | |

- 2
- | | | | |
|---|-----------|---|---------------|
| a | 2.718 280 | b | 2.718 281 828 |
|---|-----------|---|---------------|

- 3
- | | | | | | |
|---|-----------|---|-----------|---|-----------|
| a | 2.012 171 | b | 2.013 753 | c | 0.001 582 |
|---|-----------|---|-----------|---|-----------|

Exponential expressions

- 4
- | | | | | | | | |
|---|-------------------|---|-------------------|---|------------|---|------------------------------|
| a | $7e^x$ | b | $9e^x$ | c | $3e^x$ | d | e^x |
| e | $12e^{2x} + 2e^x$ | f | $-5e^{2x} - 8e^x$ | g | e^{x+2} | h | e^{5x} |
| i | e^{5x} | j | e^{7x} | k | $2e^x - 3$ | l | $\frac{e^{3x} + 7e^{2x}}{6}$ |

- 5
- | | | | |
|---|-------------------------|---|------------------------------|
| a | $e^{2x} + 3e^x$ | b | $e^{3x} - 9e^{2x}$ |
| c | $e^{2x} + 7e^x + 10$ | d | $e^{2x} - 3e^x - 10$ |
| e | $2 + 2e^x$ | f | $3 + 3e^{2x}$ |
| g | $2e + e^{2x} + e^{-2x}$ | h | $e^{2x} - 2e^{-x} + e^{-4x}$ |

- 6
- | | | | |
|---|----------------------|---|----------------------------|
| a | $4(2e^x + 3)$ | b | $4e^x(2e^x + 3)$ |
| c | $4e^{3x}(2e^x - 3)$ | d | $e^{4x}(8 - e^{3x})$ |
| e | $(e^x - 3)(e^x - 2)$ | f | $(e^x + 3)(e^x + 3)$ |
| g | $(e^x + 4)(e^x + 2)$ | h | $(e^{2x} - 5)(e^{2x} - 2)$ |



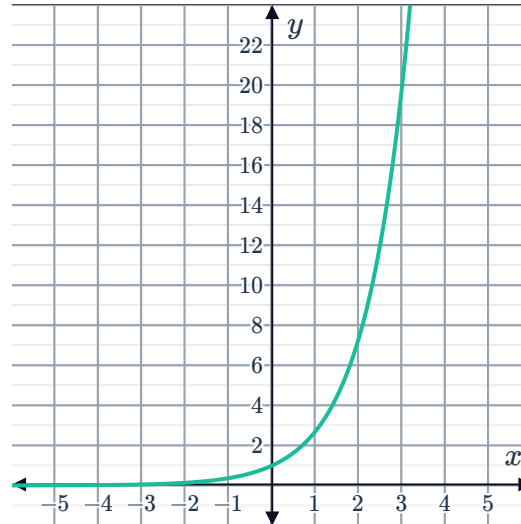
Exponential graphs

7

a

x	-3	-2	-1	0	1	2	3
$f(x)$	0.05	0.14	0.37	1.00	2.72	7.39	20.09

b



8

- a For $x > 0$, the graph of $y = e^x$ will lie above the graph of $y = (2)^x$ and below the graph of $y = (3)^x$.
- b For $x < 0$, the graph of $y = e^x$ will lie above the graph of $y = (3)^x$ and below the graph of $y = (2)^x$.

9

a No

b

i 0 ii ∞

c No

d 1

e 0

f





10 (0, 1)

11

a

x	$f(x)$
-5	-0.007
-4	-0.018
-3	-0.050
-2	-0.135
-1	-0.368
0	-1
1	-2.718
2	-7.389
3	-20.086
4	-54.598
5	-148.413
10	-22 026.5

b No

c No

d Decreasing

e As x increases, the function decreases at a faster and faster rate.

Transformations of exponential graphs

12

a

i No ii No iii No

b

i No ii Yes iii Yes

c

i Yes ii Yes iii Yes

d

i No ii Yes iii Yes

13

a Increasing

b Decreasing

c Decreasing

d Increasing

